confidence_performance_perception_uncertainty_score_SCR

August 4, 2026

1 Multilevel Analysis for Retrospective Confidence based on Skin Conductance Response, Perceived Performance Perception, and Team Gameplay Score

by Elvis Ortega-Ochoa

[8]:
```
import sys
import subprocess

subprocess.check_call([sys.executable, "-m", "pip", "install", "--upgrade",␣
 ↪"pip", "ipywidgets", "jupyterlab_widgets"])
subprocess.check_call([
    sys.executable, "-m", "pip", "install",
    "pymer4",
    "rpy2",
    "polars",
    "great-tables",
    "joblib",
    "scikit-learn",
    "pyarrow",
    "bambi",
    "arviz",
    "matplotlib",
    "pandas",
    "numpy",
    "statsmodels",
    "openpyxl",
])
```

```
Requirement already satisfied: pip in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(26.2)
Requirement already satisfied: ipywidgets in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(8.1.8)
Requirement already satisfied: jupyterlab_widgets in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.0.16)
```

```
Requirement already satisfied: comm>=0.1.3 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipywidgets) (0.2.2)
Requirement already satisfied: ipython>=6.1.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipywidgets) (9.2.0)
Requirement already satisfied: traitlets>=4.3.1 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipywidgets) (5.14.3)
Requirement already satisfied: widgetsnbextension~=4.0.14 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from ipywidgets) (4.0.15)
Requirement already satisfied: decorator in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (5.2.1)
Requirement already satisfied: ipython-pygments-lexers in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (1.1.1)
Requirement already satisfied: jedi>=0.16 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (0.19.2)
Requirement already satisfied: matplotlib-inline in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (0.1.7)
Requirement already satisfied: pexpect>4.3 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (3.0.51)
Requirement already satisfied: pygments>=2.4.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (2.19.1)
Requirement already satisfied: stack_data in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
ipython>=6.1.0->ipywidgets) (0.6.3)
Requirement already satisfied: wcwidth in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
prompt_toolkit<3.1.0,>=3.0.41->ipython>=6.1.0->ipywidgets) (0.2.13)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
jedi>=0.16->ipython>=6.1.0->ipywidgets) (0.8.4)
Requirement already satisfied: ptyprocess>=0.5 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
pexpect>4.3->ipython>=6.1.0->ipywidgets) (0.7.0)
Requirement already satisfied: executing>=1.2.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
stack_data->ipython>=6.1.0->ipywidgets) (2.2.0)
```

```
Requirement already satisfied: asttokens>=2.1.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
stack_data->ipython>=6.1.0->ipywidgets) (3.0.0)
Requirement already satisfied: pure-eval in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
stack_data->ipython>=6.1.0->ipywidgets) (0.2.3)
Requirement already satisfied: pymer4 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(0.9.2)
Requirement already satisfied: rpy2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.6.7)
Requirement already satisfied: polars in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(1.43.2)
Requirement already satisfied: great-tables in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(0.23.0)
Requirement already satisfied: joblib in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(1.5.3)
Requirement already satisfied: scikit-learn in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(1.9.0)
Requirement already satisfied: pyarrow in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(25.0.0)
Requirement already satisfied: bambi in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(0.19.0)
Requirement already satisfied: arviz in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(1.2.0)
Requirement already satisfied: matplotlib in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.10.3)
Requirement already satisfied: pandas in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.0.5)
Requirement already satisfied: numpy in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(2.2.6)
Requirement already satisfied: statsmodels in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(0.14.6)
Requirement already satisfied: openpyxl in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(3.1.5)
```

```
Requirement already satisfied: rpy2-rinterface>=3.6.6 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rpy2) (3.6.6)
Requirement already satisfied: rpy2-robjects>=3.6.5 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rpy2) (3.6.5)
Requirement already satisfied: polars-runtime-32==1.43.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from polars) (1.43.2)
Requirement already satisfied: multimark>=0.1.3 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (0.2.0)
Requirement already satisfied: faicons>=0.2.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (0.2.2)
Requirement already satisfied: htmltools>=0.4.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (0.7.0)
Requirement already satisfied: importlib-metadata in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (9.0.0)
Requirement already satisfied: nokap>=0.1.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (0.1.0)
Requirement already satisfied: typing_extensions>=3.10.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (4.16.0)
Requirement already satisfied: Babel>=2.13.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (2.18.0)
Requirement already satisfied: importlib-resources in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from great-tables) (7.1.0)
Requirement already satisfied: scipy>=1.10.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from scikit-learn) (1.18.0)
Requirement already satisfied: narwhals>=2.0.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from scikit-learn) (2.24.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from scikit-learn) (3.6.0)
Requirement already satisfied: arviz_plots<2.0.0,>=1.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (1.2.0)
Requirement already satisfied: formulae<0.7.0,>=0.6.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (0.6.2)
```

```
Requirement already satisfied: graphviz in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (0.21)
Requirement already satisfied: pytensor<4.0.0,>=3.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (3.2.4)
Requirement already satisfied: pymc<7.0.0,>=6.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (6.2.0)
Requirement already satisfied: seaborn<0.14,>=0.13.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (0.13.2)
Requirement already satisfied: sparse<0.18,>=0.17.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from bambi) (0.17.0)
Requirement already satisfied: arviz-base<1.3,>=1.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz_plots<2.0.0,>=1.0.0->bambi) (1.2.0)
Requirement already satisfied: arviz-stats<1.3,>=1.2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz-stats[xarray]<1.3,>=1.2->arviz_plots<2.0.0,>=1.0.0->bambi) (1.2.0)
Requirement already satisfied: xarray>=2024.11.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz-base<1.3,>=1.2->arviz_plots<2.0.0,>=1.0.0->bambi) (2026.7.0)
Requirement already satisfied: lazy_loader>=0.4 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz-base<1.3,>=1.2->arviz_plots<2.0.0,>=1.0.0->bambi) (0.5)
Requirement already satisfied: xarray-einstats in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from arviz-stats[xarray]<1.3,>=1.2->arviz_plots<2.0.0,>=1.0.0->bambi) (0.11.0)
Requirement already satisfied: packaging in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
formulae<0.7.0,>=0.6.0->bambi) (25.0)
Requirement already satisfied: cachetools<7,>=4.2.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pymc<7.0.0,>=6.0.0->bambi) (6.2.6)
Requirement already satisfied: cloudpickle in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pymc<7.0.0,>=6.0.0->bambi) (3.1.2)
Requirement already satisfied: rich>=13.7.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pymc<7.0.0,>=6.0.0->bambi) (15.0.0)
Requirement already satisfied: setuptools>=59.0.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pytensor<4.0.0,>=3.0.0->bambi) (83.0.0)
Requirement already satisfied: numba<=0.66.0,>=0.58 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pytensor<4.0.0,>=3.0.0->bambi) (0.66.0)
```

```
Requirement already satisfied: filelock>=3.15 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from pytensor<4.0.0,>=3.0.0->bambi) (3.32.2)
Requirement already satisfied: llvmlite<0.49,>=0.48.0dev0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from numba<=0.66.0,>=0.58->pytensor<4.0.0,>=3.0.0->bambi) (0.48.0)
Requirement already satisfied: contourpy>=1.0.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (1.3.2)
Requirement already satisfied: cycler>=0.10 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (4.58.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (1.4.8)
Requirement already satisfied: pillow>=8 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (11.2.1)
Requirement already satisfied: pyparsing>=2.3.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from matplotlib) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
matplotlib) (2.9.0.post0)
Requirement already satisfied: patsy>=0.5.6 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from statsmodels) (1.0.2)
Requirement already satisfied: et-xmlfile in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from openpyxl) (2.0.0)
Requirement already satisfied: cffi>=1.15 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from multimark>=0.1.3->great-tables) (2.1.0)
Requirement already satisfied: click>=8.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from multimark>=0.1.3->great-tables) (8.4.2)
Requirement already satisfied: pycparser in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from cffi>=1.15->multimark>=0.1.3->great-tables) (3.0)
Requirement already satisfied: websockets>=12.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from nokap>=0.1.0->great-tables) (17.0.1)
Requirement already satisfied: six>=1.5 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from python-
dateutil>=2.7->matplotlib) (1.17.0)
```

```
Requirement already satisfied: markdown-it-py>=2.2.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rich>=13.7.1->pymc<7.0.0,>=6.0.0->bambi) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/Users/elvisortegaec/Library/Python/3.13/lib/python/site-packages (from
rich>=13.7.1->pymc<7.0.0,>=6.0.0->bambi) (2.19.1)
Requirement already satisfied: mdurl~=0.1 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from markdown-it-py>=2.2.0->rich>=13.7.1->pymc<7.0.0,>=6.0.0->bambi) (0.1.2)
Requirement already satisfied: jinja2 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rpy2-robjects>=3.6.5->rpy2) (3.1.6)
Requirement already satisfied: tzlocal in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from rpy2-robjects>=3.6.5->rpy2) (5.4.4)
Requirement already satisfied: zipp>=3.20 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from importlib-metadata->great-tables) (4.1.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages
(from jinja2->rpy2-robjects>=3.6.5->rpy2) (3.0.3)
```

[8]:
```
0
```

Packages

[9]:
```
import pandas as pd
import numpy as np
import bambi as bmb
import arviz as az
import matplotlib.pyplot as plt
import matplotlib.patches as patches
import statsmodels.api as sm
import pymer4.models as pm
import polars as pl
import ipywidgets as widgets
from pymer4.models import lmer as Lmer
```

```
Error importing in API mode: ImportError("dlopen(/Library/Frameworks/Python.fram
ework/Versions/3.13/lib/python3.13/site-packages/_rinterface_cffi_api.abi3.so,
0x0002): Library not loaded: /Library/Frameworks/R.framework/Versions/4.5-arm64/
Resources/lib/libRblas.dylib\n  Referenced from:
<2F94BB47-26EE-3EB3-85C5-1EA8A83C71AB>
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/_rinterface_cffi_api.abi3.so\n  Reason: tried: '/Library/Frameworks/R.f
ramework/Versions/4.5-arm64/Resources/lib/libRblas.dylib' (no such file), '/Syst
em/Volumes/Preboot/Cryptexes/OS/Library/Frameworks/R.framework/Versions/4.5-arm6
4/Resources/lib/libRblas.dylib' (no such file), '/Library/Frameworks/R.framework
/Versions/4.5-arm64/Resources/lib/libRblas.dylib' (no such file)")
```

```
Trying to import in ABI mode.
```

Loading data

[10]:
```
# Reading files and printing columns
scr = pd.read_excel("../../confidence_SCR/stable_scr_events_UPD.xlsx",␣
 ↪keep_default_na=False)
cf = pd.read_excel("../../confidence_SCR/CONT_dependent_variables_confidence_E.
 ↪xlsx", keep_default_na=False)

# Make a clean copy
scr = scr.copy()
cf = cf.copy()

# Extract participantID and sensorID from scrID
scr["participantID"] = scr["scrID"].str.extract(r"SCR_([^.]+)\.")
scr["sensorID"] = scr["scrID"].str.extract(r"\.([A-Z]{2})-EDA$")

# Merge the dataframes on groupID, participantID, and sensorID
dataset = scr.merge(
    cf,
    on=["groupID", "participantID", "sensorID"],
    how="inner"
)

# Rename columns for clarity
dataset = dataset.rename(columns={
    "realGamePeaks/m": "realGamePeaks_per_m",
    "realTutoPeaks/m": "realTutoPeaks_per_m",
    "realQuestionsPeaks/m": "realQuestionsPeaks_per_m",
    "realPitchPeaks/m": "realPitchPeaks_per_m",
    "CF1": "cf",
    "PP1": "pp"
})
```

Build questionPath as numeric

[11]:
```
# Build numeric questionPath at group level
dataset["qBonusPeaks"] = pd.to_numeric(dataset["qBonusPeaks"], errors="coerce")
dataset["qPenaltyPeaks"] = pd.to_numeric(dataset["qPenaltyPeaks"],␣
 ↪errors="coerce")

dataset["has_bonus"] = dataset["qBonusPeaks"].notna()
dataset["has_penalty"] = dataset["qPenaltyPeaks"].notna()

group_path = (
    dataset.groupby("groupID")[["has_bonus", "has_penalty"]]
    .any()
```

```
    .reset_index()
)

group_path["questionPath"] = np.select(
    [
        group_path["has_bonus"] & ~group_path["has_penalty"],
        group_path["has_penalty"] & ~group_path["has_bonus"],
    ],
    [1, 0],
    default=np.nan,
)

# Drop any existing questionPath variants to avoid suffixes, then merge
dataset = dataset.drop(columns=[c for c in dataset.columns if c.
 ↪startswith("questionPath")], errors="ignore")
dataset = dataset.merge(group_path[["groupID", "questionPath"]], on="groupID",␣
 ↪how="left")
```

Reverse team gameplay score

[12]:
```
# Build numeric uncertScore
dataset["uncertScore"] = pd.to_numeric(dataset["uncertScore"], errors="coerce")

# Reverse-coding, using the instrument's theoretical bounds, not the sample's
SCALE_MIN, SCALE_MAX = 1, 70  # instrument range, per ¶102
dataset["uncertScore"] = (SCALE_MIN + SCALE_MAX) - dataset["uncertScore"]

# Reverse-score uncertScore using the observed scale range
#dataset["uncertScore"] = dataset["uncertScore"].max() + dataset["uncertScore"].
 ↪min() - dataset["uncertScore"]
```

[13]:
```
# Finalize data types
dataset["groupID"] = dataset["groupID"].astype("category")
dataset["cf"] = pd.to_numeric(dataset["cf"], errors="coerce")
dataset["questionPath"] = pd.Categorical(dataset["questionPath"],␣
 ↪categories=[0,1])
```

[14]:
```
# Keep just variables needed
dat_model = dataset[["groupID", "participantID", "cf", "pp", "uncertScore",␣
 ↪"realGamePeaks_per_m", "realTutoPeaks_per_m", "realQuestionsPeaks_per_m",␣
 ↪"realPitchPeaks_per_m", "questionPath", "pitchRole"]].copy()
```

1.1 Mixed Linear Models

1.1.1 Within/Between Descomposition

[15]:
```
def add_within_between(df, raw_col, prefix):
    group_mean = df.groupby("groupID")[raw_col].transform("mean")
    within = df[raw_col] - group_mean
    between = group_mean - group_mean.mean()

    df[f"{prefix}_group_mean"] = group_mean
    df[f"{prefix}_within"] = within
    df[f"{prefix}_between"] = between

    return df

dat_model = add_within_between(dat_model, "realGamePeaks_per_m", "scr")
dat_model = add_within_between(dat_model, "realTutoPeaks_per_m", "tuto")
dat_model = add_within_between(dat_model, "realQuestionsPeaks_per_m",␣
 ↪"questions")
dat_model = add_within_between(dat_model, "realPitchPeaks_per_m", "pitch")
dat_model = add_within_between(dat_model, "pp", "pp")
```

1.1.2 Standarize predictor variables

[16]:
```
dat_model["realGamePeaks_per_m_z"] = (
    dat_model["realGamePeaks_per_m"] - dat_model["realGamePeaks_per_m"].mean()
) / dat_model["realGamePeaks_per_m"].std()

dat_model["scr_within_z"] = (
    dat_model["scr_within"] - dat_model["scr_within"].mean()
) / dat_model["scr_within"].std()

dat_model["scr_between_z"] = (
    dat_model["scr_between"] - dat_model["scr_between"].mean()
) / dat_model["scr_between"].std()

dat_model["tuto_within_z"] = (
    dat_model["tuto_within"] - dat_model["tuto_within"].mean()
) / dat_model["tuto_within"].std()

dat_model["tuto_between_z"] = (
    dat_model["tuto_between"] - dat_model["tuto_between"].mean()
) / dat_model["tuto_between"].std()

dat_model["questions_within_z"] = (
    dat_model["questions_within"] - dat_model["questions_within"].mean()
) / dat_model["questions_within"].std()
```

```
dat_model["questions_between_z"] = (
    dat_model["questions_between"] - dat_model["questions_between"].mean()
) / dat_model["questions_between"].std()

dat_model["pitch_within_z"] = (
    dat_model["pitch_within"] - dat_model["pitch_within"].mean()
) / dat_model["pitch_within"].std()

dat_model["pitch_between_z"] = (
    dat_model["pitch_between"] - dat_model["pitch_between"].mean()
) / dat_model["pitch_between"].std()

dat_model["pp_z"] = (
    dat_model["pp"] - dat_model["pp"].mean()
) / dat_model["pp"].std()

dat_model["pp_within_z"] = (
    dat_model["pp_within"] - dat_model["pp_within"].mean()
) / dat_model["pp_within"].std()

dat_model["pp_between_z"] = (
    dat_model["pp_between"] - dat_model["pp_between"].mean()
) / dat_model["pp_between"].std()

dat_model["uncertScore_z"] = (
    dat_model["uncertScore"] - dat_model["uncertScore"].mean()
) / dat_model["uncertScore"].std()

dat_model.columns.tolist()
```

[16]:
```
['groupID',
 'participantID',
 'cf',
 'pp',
 'uncertScore',
 'realGamePeaks_per_m',
 'realTutoPeaks_per_m',
 'realQuestionsPeaks_per_m',
 'realPitchPeaks_per_m',
 'questionPath',
 'pitchRole',
 'scr_group_mean',
 'scr_within',
 'scr_between',
 'tuto_group_mean',
 'tuto_within',
 'tuto_between',
```

```
 'questions_group_mean',
 'questions_within',
 'questions_between',
 'pitch_group_mean',
 'pitch_within',
 'pitch_between',
 'pp_group_mean',
 'pp_within',
 'pp_between',
 'realGamePeaks_per_m_z',
 'scr_within_z',
 'scr_between_z',
 'tuto_within_z',
 'tuto_between_z',
 'questions_within_z',
 'questions_between_z',
 'pitch_within_z',
 'pitch_between_z',
 'pp_z',
 'pp_within_z',
 'pp_between_z',
 'uncertScore_z']
```

Pitch data cleaning

[17]:
```
dat_model['pitchRole'] = pd.to_numeric(dat_model['pitchRole'], errors='coerce')
dat_model_pitch_role = dat_model.dropna(subset=['groupID',
 'participantID',
 'cf',
 'pp',
 'uncertScore',
 'realGamePeaks_per_m',
 'realTutoPeaks_per_m',
 'realQuestionsPeaks_per_m',
 'realPitchPeaks_per_m',
 'questionPath',
 'pitchRole',
 'scr_group_mean',
 'scr_within',
 'scr_between',
 'tuto_group_mean',
 'tuto_within',
 'tuto_between',
 'questions_group_mean',
 'questions_within',
 'questions_between',
 'pitch_group_mean',
```

```
 'pitch_within',
 'pitch_between',
 'pp_group_mean',
 'pp_within',
 'pp_between',
 'realGamePeaks_per_m_z',
 'scr_within_z',
 'scr_between_z',
 'tuto_within_z',
 'tuto_between_z',
 'questions_within_z',
 'questions_between_z',
 'pitch_within_z',
 'pitch_between_z',
 'pp_z',
 'pp_within_z',
 'pp_between_z',
 'uncertScore_z']).copy()

dat_model_pitch_role = dat_model_pitch_role[dat_model_pitch_role['pitchRole'].
 ↪isin([0, 1])].copy()

# Finalize data type
dat_model_pitch_role["pitchRole"] = pd.
 ↪Categorical(dat_model_pitch_role["pitchRole"], categories=[0, 1])
```

1.2 Exploratory Data Analysis

[18]:
```
cols = ["cf", "pp", "uncertScore", "realGamePeaks_per_m",␣
 ↪"realTutoPeaks_per_m", "realQuestionsPeaks_per_m", "realPitchPeaks_per_m"]

# Basic summary
print("Missing values:")
print(dat_model[cols].isna().sum(), "\n")

print("Descriptive statistics:")
display(dat_model[cols].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols), figsize=(20, 8))

for i, col in enumerate(cols):
    axes[0, i].hist(dat_model[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")
```

```
    axes[1, i].boxplot(dat_model[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

plt.tight_layout()
plt.show()

# Correlation matrix
corr = dat_model[cols].corr()

fig, ax = plt.subplots(figsize=(8, 6))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

ax.set_xticks(range(len(cols)))
ax.set_yticks(range(len(cols)))
ax.set_xticklabels(cols, rotation=45, ha="right")
ax.set_yticklabels(cols)
ax.set_title("General Correlation Matrix")

# Add correlation scores in each cell
for i in range(len(cols)):
    for j in range(len(cols)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",␣
 ↪color=text_color)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```

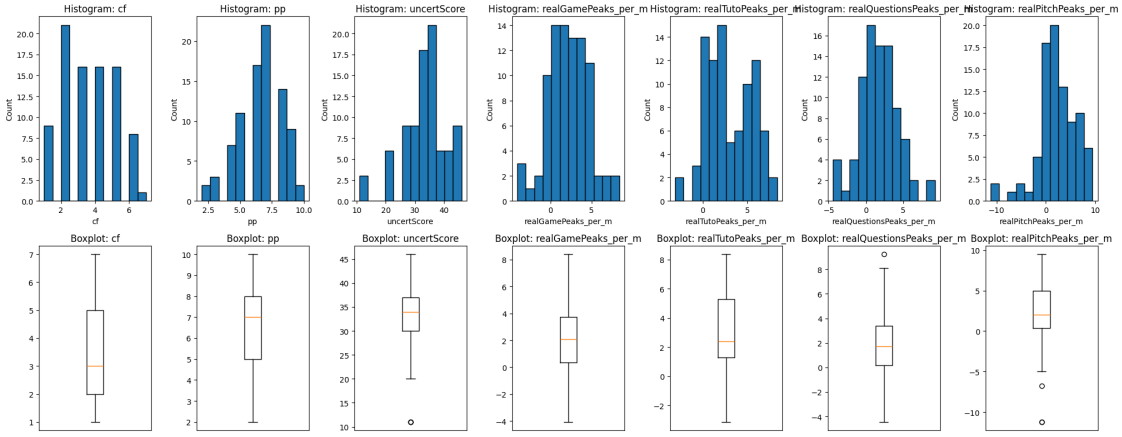
```
Missing values:
cf                          0
pp                          0
uncertScore                 0
realGamePeaks_per_m         0
realTutoPeaks_per_m         0
realQuestionsPeaks_per_m    0
realPitchPeaks_per_m        0
dtype: int64

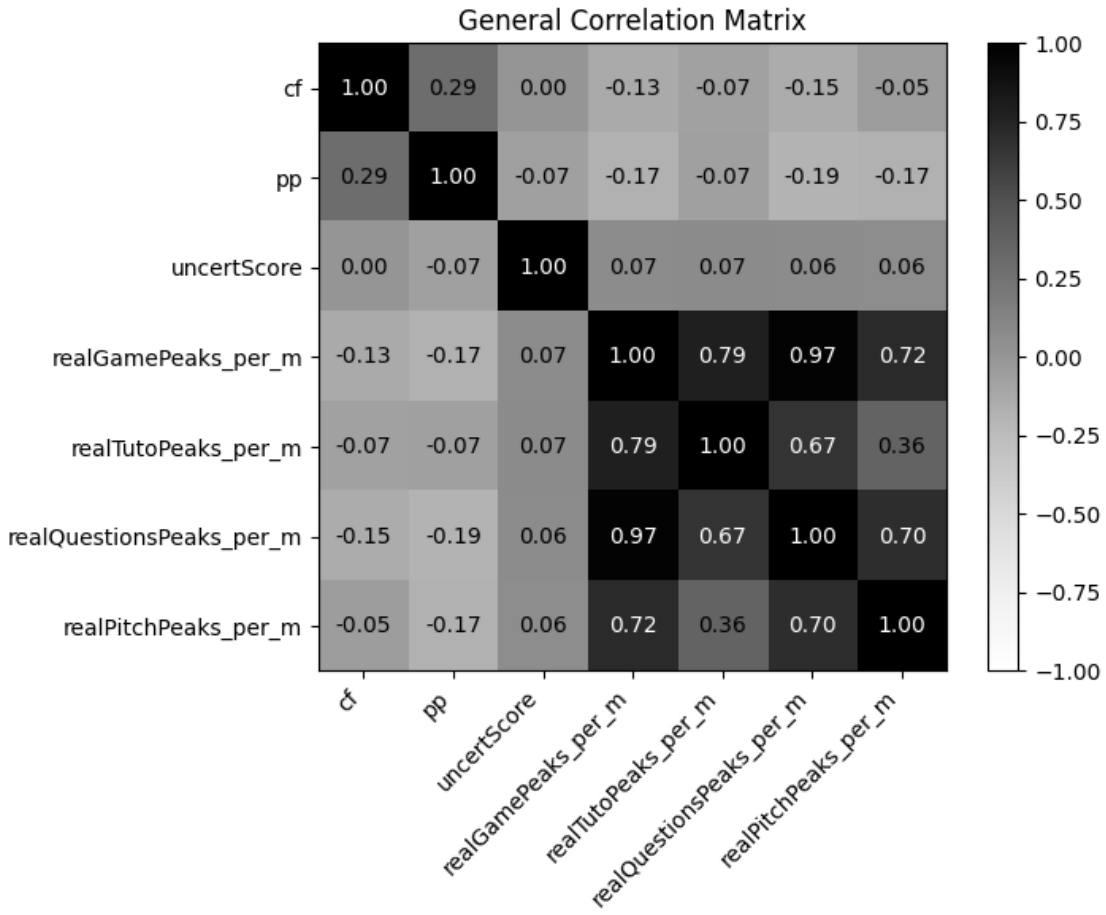
Descriptive statistics:

                          count       mean       std        min        25%  \
cf                         87.0   3.425287  1.559906   1.000000   2.000000
pp                         87.0   6.494253  1.764466   2.000000   5.000000
uncertScore                87.0  33.482759  7.603338  11.000000  30.000000
realGamePeaks_per_m        87.0   2.114204  2.406057  -4.070671   0.358493
```

realTutoPeaks_per_m	87.0	2.989865	2.548684	-3.124891	1.270447
realQuestionsPeaks_per_m	87.0	1.800861	2.490374	-4.446577	0.189259
realPitchPeaks_per_m	87.0	2.272550	3.905028	-11.221374	0.346963

	50%	75%	max
cf	3.000000	5.000000	7.000000
pp	7.000000	8.000000	10.000000
uncertScore	34.000000	37.000000	46.000000
realGamePeaks_per_m	2.079011	3.722039	8.403172
realTutoPeaks_per_m	2.378785	5.282077	8.361345
realQuestionsPeaks_per_m	1.723939	3.370871	9.234457
realPitchPeaks_per_m	2.019637	4.973874	9.502378





[19]:
```
eda_df = dat_model.copy()

eda_df[["pitch_within", "pitch_between"]] = np.nan
eda_df.loc[dat_model_pitch_role.index, ["pitch_within", "pitch_between"]] =␣
 ↪dat_model_pitch_role[["pitch_within", "pitch_between"]].to_numpy()

cols_within_between = ["cf", "pp_within", "pp_between", "uncertScore",␣
 ↪"scr_within", "scr_between", "tuto_within",␣
 ↪"tuto_between","questions_within", "questions_between", "pitch_within",␣
 ↪"pitch_between"]
cols_within = ["cf", "pp_within", "uncertScore", "scr_within", "tuto_within",␣
 ↪"questions_within", "pitch_within"]
cols_between = ["cf", "pp_between", "uncertScore", "scr_between", ␣
 ↪"tuto_between", "questions_between", "pitch_between"]
```

Within-team Exploratory Data Analysis

[20]:
```
# Basic summary
print("Missing values:")
print(eda_df[cols_within].isna().sum(), "\n")

print("Descriptive statistics:")
display(eda_df[cols_within].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols_within), figsize=(20, 8))

for i, col in enumerate(cols_within):
    axes[0, i].hist(eda_df[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(eda_df[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

plt.tight_layout()
plt.show()
```

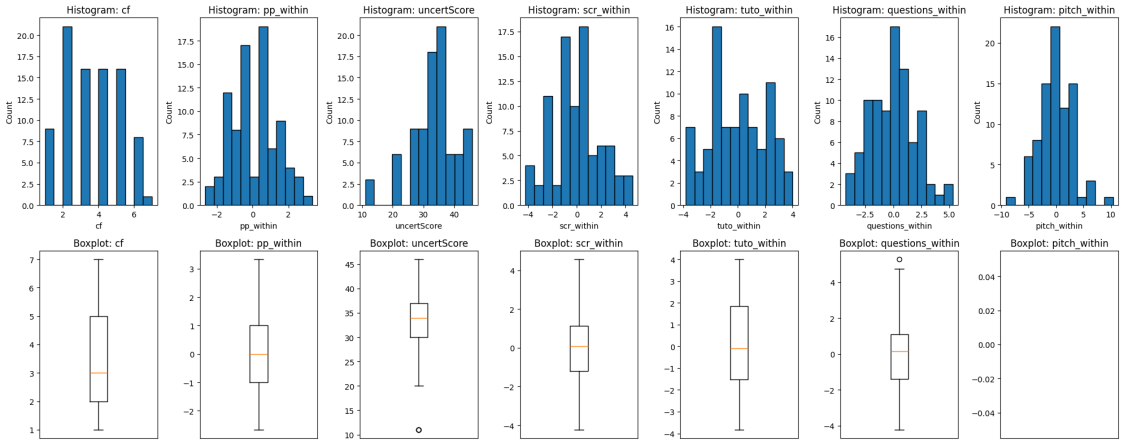
```
Missing values:
cf                  0
pp_within           0
uncertScore         0
scr_within          0
tuto_within         0
questions_within    0
pitch_within        3
dtype: int64

Descriptive statistics:
```

	count	mean	std	min	25%	\
cf	87.0	3.425287e+00	1.559906	1.000000	2.000000	
pp_within	87.0	-5.104474e-17	1.275895	-2.666667	-1.000000	
uncertScore	87.0	3.348276e+01	7.603338	11.000000	30.000000	
scr_within	87.0	4.083579e-17	1.952139	-4.228912	-1.186685	
tuto_within	87.0	1.020895e-17	2.049427	-3.827188	-1.513568	
questions_within	87.0	-1.786566e-17	1.992601	-4.223753	-1.400472	
pitch_within	84.0	-1.057355e-17	3.098589	-9.181356	-1.706938	

	50%	75%	max
cf	3.000000	5.000000	7.000000
pp_within	0.000000	1.000000	3.333333
uncertScore	34.000000	37.000000	46.000000

```
scr_within          0.079387   1.136715   4.572771
tuto_within        -0.096863   1.860652   4.003711
questions_within    0.153288   1.088501   5.281570
pitch_within       -0.176149   2.024036  10.406666
```



[21]:
```python
# Correlation matrix
corr = eda_df[cols_within].corr()

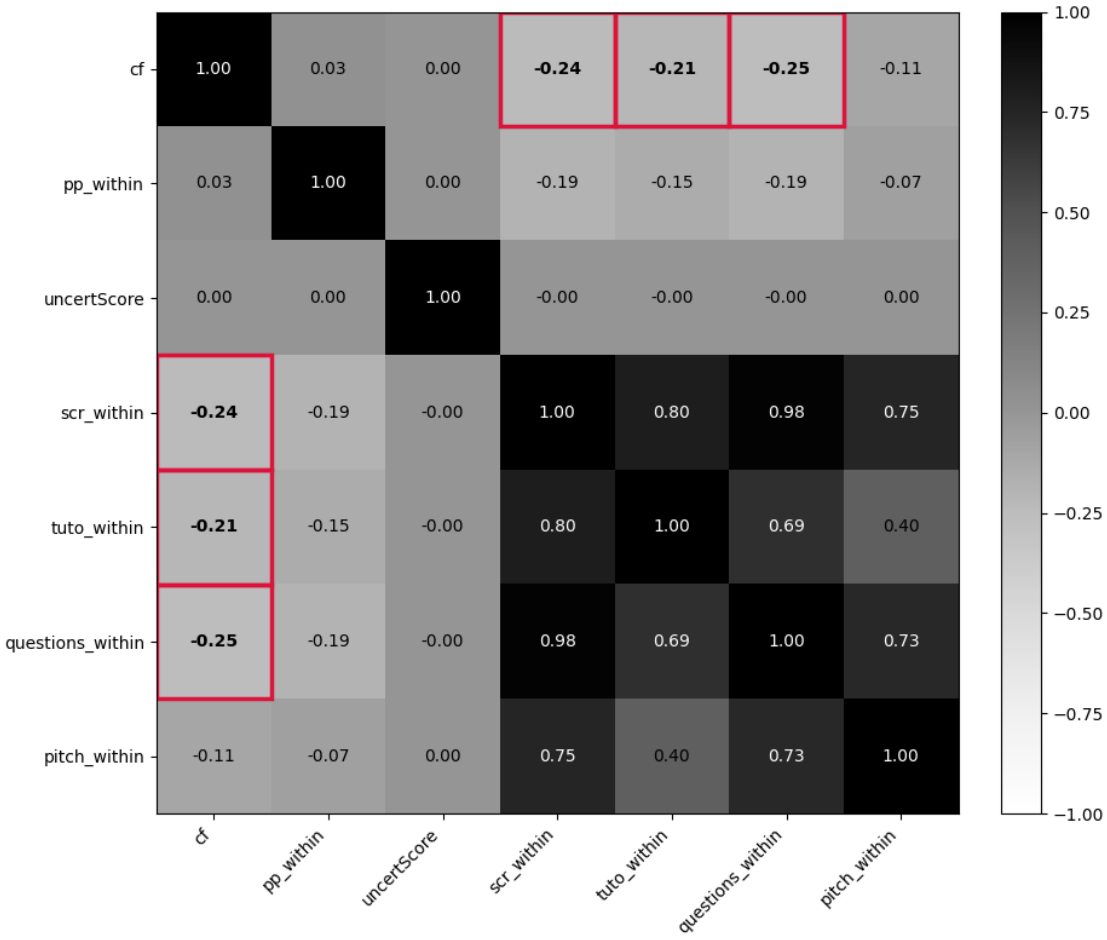
fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

ax.set_xticks(range(len(cols_within)))
ax.set_yticks(range(len(cols_within)))
ax.set_xticklabels(cols_within, rotation=45, ha="right")
ax.set_yticklabels(cols_within)

threshold = 0.20
focus_var = "cf"

for i in range(len(cols_within)):
    for j in range(len(cols_within)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_within[i], cols_within[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)
```

```
plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



Between-team Exploratory Data Analysis

[22]:
```
# Basic summary
print("Missing values:")
print(eda_df[cols_between].isna().sum(), "\n")

print("Descriptive statistics:")
display(eda_df[cols_between].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols_between), figsize=(20, 8))

for i, col in enumerate(cols_between):
```

```
    axes[0, i].hist(eda_df[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(eda_df[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

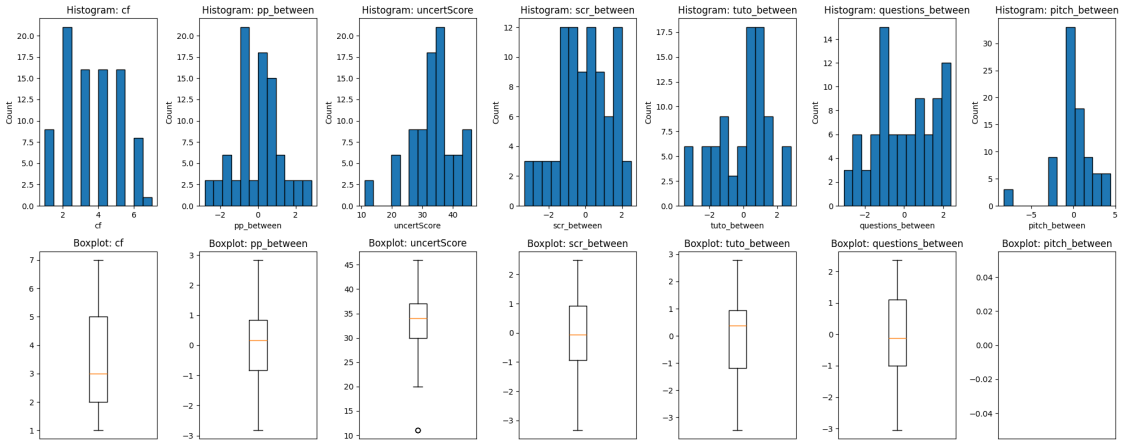
plt.tight_layout()
plt.show()
```

```
Missing values:
cf                  0
pp_between          0
uncertScore         0
scr_between         0
tuto_between        0
questions_between   0
pitch_between       3
dtype: int64

Descriptive statistics:
```

	count	mean	std	min	25%
cf	87.0	3.425287e+00	1.559906	1.000000	2.000000
pp_between	87.0	-1.327163e-16	1.218783	-2.827586	-0.827586
uncertScore	87.0	3.348276e+01	7.603338	11.000000	30.000000
scr_between	87.0	2.756416e-16	1.406507	-3.337750	-0.941966
tuto_between	87.0	-8.473426e-16	1.515138	-3.478591	-1.187725
questions_between	87.0	8.167158e-17	1.493821	-3.056665	-1.009198
pitch_between	84.0	1.269501e-01	2.385979	-8.290591	-0.652798

	50%	75%	max
cf	3.000000	5.000000	7.000000
pp_between	0.172414	0.839080	2.839080
uncertScore	34.000000	37.000000	46.000000
scr_between	-0.065864	0.914374	2.494181
tuto_between	0.369232	0.947484	2.792636
questions_between	-0.114830	1.106064	2.363263
pitch_between	0.139819	1.189454	4.415574



[23]:
```
# Correlation matrix
corr = eda_df[cols_between].corr()

fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

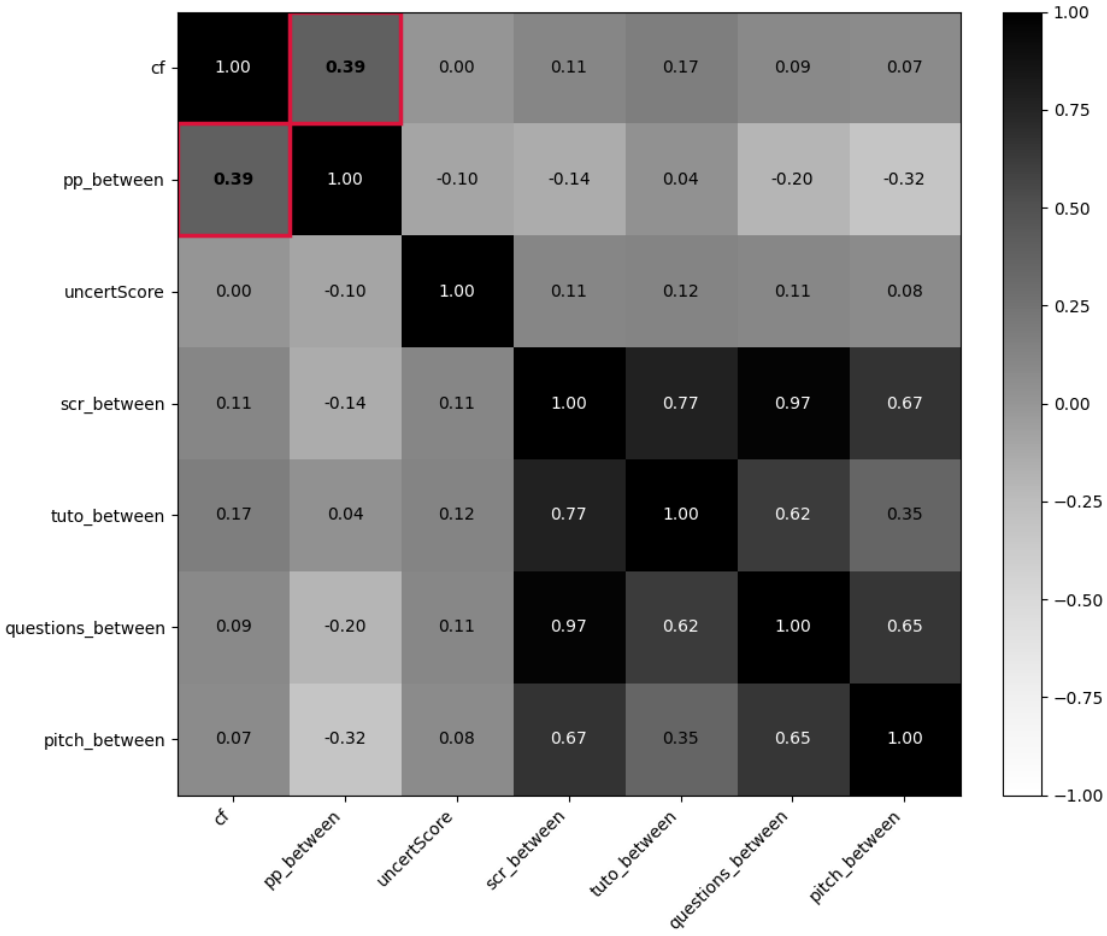
ax.set_xticks(range(len(cols_between)))
ax.set_yticks(range(len(cols_between)))
ax.set_xticklabels(cols_between, rotation=45, ha="right")
ax.set_yticklabels(cols_between)
#ax.set_title("Detailed Correlation Matrix")

threshold = 0.20
focus_var = "cf"

for i in range(len(cols_between)):
    for j in range(len(cols_between)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_between[i], cols_between[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
```

```
plt.show()
```



[24]:
```
# Correlation matrix
corr = eda_df[cols_within_between].corr()

fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

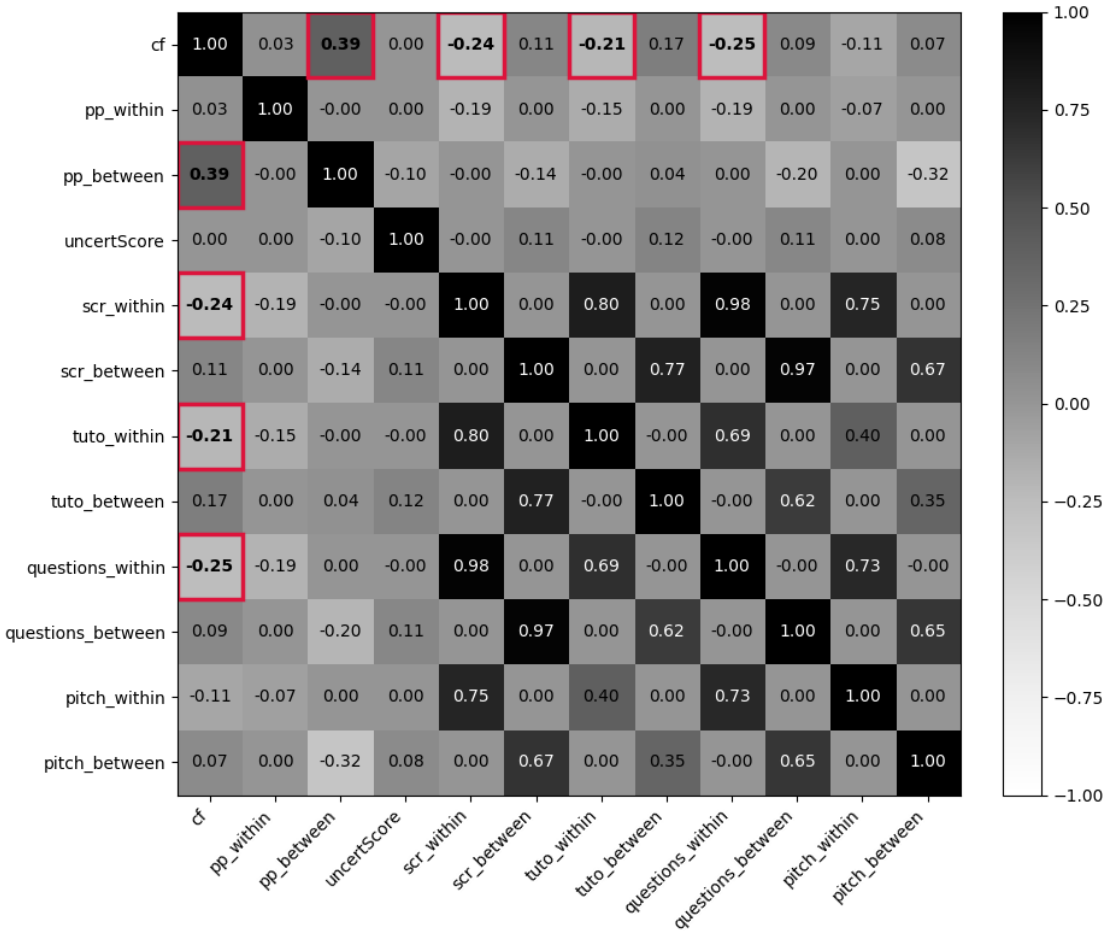
ax.set_xticks(range(len(cols_within_between)))
ax.set_yticks(range(len(cols_within_between)))
ax.set_xticklabels(cols_within_between, rotation=45, ha="right")
ax.set_yticklabels(cols_within_between)

threshold = 0.20
focus_var = "cf"

for i in range(len(cols_within_between)):
```

```
    for j in range(len(cols_within_between)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_within_between[i],␣
 ↪cols_within_between[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



1.2.1 Model 1 - Game play SCR: CF ~ realGamePeaks_per_m_z + (1|groupID)

1. Statsmodels (LME4)

[25]:
```
model1_sm = sm.MixedLM.from_formula(
    "cf ~ realGamePeaks_per_m_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_sm = model1_sm.fit(reml=True, method="lbfgs")
print("Model 1 sm: CF ~ realGamePeaks_per_m_z + (1|groupID)")
print(result1_sm.summary())
```

```
Model 1 sm: CF ~ realGamePeaks_per_m_z + (1|groupID)
             Mixed Linear Model Regression Results
================================================================
Model:               MixedLM    Dependent Variable:    cf
No. Observations:    87         Method:                REML
No. Groups:          29         Scale:                 1.8671
Min. group size:     3          Log-Likelihood:        -160.6816
Max. group size:     3          Converged:             Yes
Mean group size:     3.0
----------------------------------------------------------------
                      Coef.  Std.Err.   z    P>|z| [0.025 0.975]
----------------------------------------------------------------
Intercept              3.425    0.203 16.854 0.000  3.027  3.824
realGamePeaks_per_m_z -0.298    0.167 -1.789 0.074 -0.625  0.029
groupID Var            0.575    0.295
================================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[26]:
```
model1_bmb = bmb.Model(
    "cf ~ realGamePeaks_per_m_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1 = model1_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
```

```
    cores=4,
    random_seed=123
)

print("Model 1 bmb: CF ~ realGamePeaks_per_m_z + (1|groupID)")
az.summary(idata_model1, var_names=["realGamePeaks_per_m_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, realGamePeaks_per_m_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
```

```
Model 1 bmb: CF ~ realGamePeaks_per_m_z + (1|groupID)
```

[26]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	\
realGamePeaks_per_m_z	-0.287	0.17	-0.56	-0.013	5372	4188	
1|groupID_sigma	0.73	0.28	0.23	1.2	1606	1980	
sigma	1.405	0.138	1.2	1.6	2833	3477	

	r_hat	mcse_mean	mcse_sd
realGamePeaks_per_m_z	1.00	0.0023	0.0017
1|groupID_sigma	1.00	0.0071	0.0053
sigma	1.00	0.0026	0.0019

Within/Between Model 1: CF ~ scr_within_z + scr_between_z + (1|groupID)

1. Statsmodels (LME4)

[27]:
```
model1_within_sm = sm.MixedLM.from_formula(
    "cf ~ scr_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_within_sm = model1_within_sm.fit(reml=True, method="lbfgs")
print("Model 1 within sm: CF ~ scr_within_z + (1|groupID)")
print(result1_within_sm.summary())
```

```
Model 1 within sm: CF ~ scr_within_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: cf
No. Observations: 87       Method:             REML
```

```
No. Groups:       29      Scale:              1.8480
Min. group size:  3       Log-Likelihood:     -159.2149
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
-----------------------------------------------------
             Coef.  Std.Err.   z    P>|z| [0.025 0.975]
-----------------------------------------------------
Intercept     3.425    0.194 17.632 0.000  3.045  3.806
scr_within_z -0.374    0.147 -2.548 0.011 -0.661 -0.086
groupID Var   0.478    0.263
=====================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[28]:
```
model1_between_sm = sm.MixedLM.from_formula(
    "cf ~ scr_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_between_sm = model1_between_sm.fit(reml=True, method="lbfgs")
print("Model 1 between sm: CF ~ scr_between_z + (1|groupID)")
print(result1_between_sm.summary())
```

```
Model 1 between sm: CF ~ scr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=======================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              2.0230
Min. group size:  3       Log-Likelihood:     -161.6428
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
-------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
-------------------------------------------------------
Intercept     3.425    0.195 17.573 0.000  3.043  3.807
scr_between_z 0.177    0.196  0.901 0.368 -0.208  0.561
groupID Var   0.427    0.255
=======================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
```

```
  warnings.warn(msg)
```

[29]:
```
model1_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ scr_within_z + scr_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_within_between_sm = model1_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 1 within-between sm: CF ~ scr_within_z + scr_between_z +␣
 ↪(1|groupID)")
print(result1_within_between_sm.summary())
```

```
Model 1 within-between sm: CF ~ scr_within_z + scr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
============================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              1.8480
Min. group size:   3       Log-Likelihood:     -159.5196
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
------------------------------------------------------------
              Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept      3.425    0.195 17.573 0.000  3.043  3.807
scr_within_z  -0.374    0.147 -2.548 0.011 -0.661 -0.086
scr_between_z  0.177    0.196  0.901 0.368 -0.208  0.561
groupID Var    0.486    0.268
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[30]:
```
model1_within_bmb  = bmb.Model(
    "cf ~ scr_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_within_bmb = model1_within_bmb.fit(
    draws=2000,
    tune=2000,
```

```
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 1 within bmb: CF ~ scr_within_z + (1|groupID)")
az.summary(idata_model1_within_bmb, var_names=["scr_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, scr_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
```

```
Model 1 within bmb: CF ~ scr_within_z + (1|groupID)
```

[30]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
scr_within_z	-0.375	0.155	-0.62	-0.13	13289	5843	1.00	
1|groupID_sigma	0.66	0.269	0.18	1.1	1851	1841	1.00	
sigma	1.397	0.132	1.2	1.6	3681	5320	1.00	

	mcse_mean	mcse_sd
scr_within_z	0.0013	0.001
1|groupID_sigma	0.0063	0.0044
sigma	0.0022	0.0016

[31]:
```
model1_between_bmb  = bmb.Model(
    "cf ~ scr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_between_bmb = model1_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 1 between bmb: CF ~ scr_between_z + (1|groupID)")
az.summary(idata_model1_between_bmb, var_names=["scr_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, scr_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model 1 between bmb: CF ~ scr_between_z + (1|groupID)
```

[31]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
scr_between_z	0.173	0.201	-0.15	0.5	6226	5838	1.00	
1|groupID_sigma	0.61	0.28	0.12	1	1453	1668	1.00	
sigma	1.46	0.133	1.3	1.7	3505	4230	1.00	

	mcse_mean	mcse_sd
scr_between_z	0.0025	0.0019
1|groupID_sigma	0.0073	0.005
sigma	0.0023	0.0016

[32]:
```
model1_within_between_bmb  = bmb.Model(
    "cf ~ scr_within_z + scr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_within_between_bmb = model1_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 1 within-between bmb: CF ~ scr_within_z + scr_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model1_within_between_bmb, var_names=["scr_within_z",␣
 ↪"scr_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, scr_within_z, scr_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.

Model 1 within-between bmb: CF ~ scr_within_z + scr_between_z + (1|groupID)
```

[32]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
scr_within_z	-0.373	0.149	-0.61	-0.13	11982	6003	1.00	
scr_between_z	0.179	0.201	-0.14	0.5	5997	5746	1.00	
1|groupID_sigma	0.66	0.28	0.18	1.1	1405	1236	1.00	
sigma	1.397	0.131	1.2	1.6	2930	3775	1.00	

	mcse_mean	mcse_sd
scr_within_z	0.0014	0.00099
scr_between_z	0.0026	0.0019
1|groupID_sigma	0.0074	0.0052
sigma	0.0024	0.0018

1.2.2 Model 2 - Event Phase: CF ~ event_phase_within/between_z + (1|groupID)

a) Tutorial

1. Statsmodels (LME4)

[33]:
```
model2_tuto_within_sm = sm.MixedLM.from_formula(
    "cf ~ tuto_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_tuto_within_sm = model2_tuto_within_sm.fit(reml=True, method="lbfgs")
print("Model 2 tuto within sm: CF ~ tuto_within_z + (1|groupID)")
print(result2_tuto_within_sm.summary())
```

```
Model 2 tuto within sm: CF ~ tuto_within_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              1.8934
Min. group size:   3       Log-Likelihood:     -159.9074
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
              Coef.  Std.Err.    z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept      3.425    0.194 17.632 0.000  3.045  3.806
tuto_within_z -0.331    0.148 -2.229 0.026 -0.622 -0.040
groupID Var    0.463    0.260
=========================================================
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[34]:
```
model2_tuto_between_sm = sm.MixedLM.from_formula(
    "cf ~ tuto_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_tuto_between_sm = model2_tuto_between_sm.fit(reml=True, method="lbfgs")
print("Model 2 tuto between sm: CF ~ tuto_between_z + (1|groupID)")
print(result2_tuto_between_sm.summary())
```

```
Model 2 tuto between sm: CF ~ tuto_between_z + (1|groupID)
         Mixed Linear Model Regression Results
============================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              2.0230
Min. group size:   3       Log-Likelihood:     -161.1722
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
------------------------------------------------------------
               Coef. Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept      3.425    0.192 17.882 0.000  3.050  3.801
tuto_between_z 0.258    0.193  1.341 0.180 -0.119  0.636
groupID Var    0.390    0.246
============================================================
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[35]:
```
model2_tuto_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ tuto_within_z + tuto_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_tuto_within_between_sm = model2_tuto_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 tuto within-between sm: CF ~ tuto_within_z + tuto_between_z +␣
 ↪(1|groupID)")
```

```
print(result2_tuto_within_between_sm.summary())
```

```
Model 2 tuto within-between sm: CF ~ tuto_within_z + tuto_between_z +
(1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   87      Method:             REML
No. Groups:         29      Scale:              1.8934
Min. group size:    3       Log-Likelihood:     -159.7416
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.425    0.192 17.882 0.000  3.050  3.801
tuto_within_z  -0.331    0.148 -2.229 0.026 -0.622 -0.040
tuto_between_z  0.258    0.193  1.341 0.180 -0.119  0.636
groupID Var     0.433    0.255
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[36]:
```
model2_tuto_within_bmb  = bmb.Model(
    "cf ~ tuto_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_tuto_within_bmb = model2_tuto_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 tuto within bmb: CF ~ tuto_within_z + (1|groupID)")
az.summary(idata_model2_tuto_within_bmb, var_names=["tuto_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
```

```
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, tuto_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model 2 tuto within bmb: CF ~ tuto_within_z + (1|groupID)
```

[36]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
tuto_within_z	-0.331	0.153	-0.58	-0.086	10654	6332	1.00	
1|groupID_sigma	0.63	0.28	0.15	1.1	1306	1334	1.01	
sigma	1.419	0.137	1.2	1.7	2809	3794	1.00	

	mcse_mean	mcse_sd
tuto_within_z	0.0015	0.0011
1|groupID_sigma	0.0076	0.0053
sigma	0.0026	0.0019

[37]:
```
model2_tuto_between_bmb  = bmb.Model(
    "cf ~ tuto_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_tuto_between_bmb = model2_tuto_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 tuto between bmb: CF ~ tuto_between_z + (1|groupID)")
az.summary(idata_model2_tuto_between_bmb, var_names=["tuto_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, tuto_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

Model 2 tuto between bmb: CF ~ tuto_between_z + (1|groupID)

[37]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat
tuto_between_z	0.257	0.196	-0.053	0.57	5968	5501	1.00
1|groupID_sigma	0.58	0.283	0.1	1	1406	1442	1.00
sigma	1.458	0.134	1.3	1.7	3579	4540	1.00

	mcse_mean	mcse_sd
tuto_between_z	0.0026	0.002
1|groupID_sigma	0.0074	0.0048
sigma	0.0022	0.0016

[38]:
```
model2_tuto_within_between_bmb  = bmb.Model(
    "cf ~ tuto_within_z + tuto_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_tuto_within_between_bmb = model2_tuto_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 tuto within-between bmb: CF ~ tuto_within_z + tuto_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model2_tuto_within_between_bmb, var_names=["tuto_within_z",␣
 ↪"tuto_between_z", "1|groupID_sigma", "sigma"])
```

Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, tuto_within_z, tuto_between_z, 1|groupID_sigma, 1|groupID_offset]

Output()

Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws total) took 2 seconds.

Model 2 tuto within-between bmb: CF ~ tuto_within_z + tuto_between_z + (1|groupID)

[38]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat
tuto_within_z	-0.331	0.154	-0.58	-0.08	11870	6376	1.00
tuto_between_z	0.259	0.197	-0.054	0.58	6119	5944	1.00

```
1|groupID_sigma    0.62   0.28      0.14        1       1324       1297  1.00
sigma             1.414  0.133      1.2       1.6       3106       4349  1.00

                mcse_mean mcse_sd
tuto_within_z      0.0014   0.001
tuto_between_z     0.0025  0.0018
1|groupID_sigma    0.0075  0.0052
sigma              0.0024  0.0018
```

b) Questions

1. Statsmodels (LME4)

[39]:
```
model2_questions_within_sm = sm.MixedLM.from_formula(
    "cf ~ questions_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_questions_within_sm = model2_questions_within_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 questions within sm: CF ~ questions_within_z + (1|groupID)")
print(result2_questions_within_sm.summary())
```

```
Model 2 questions within sm: CF ~ questions_within_z + (1|groupID)
            Mixed Linear Model Regression Results
============================================================
Model:              MixedLM   Dependent Variable:   cf
No. Observations:   87        Method:               REML
No. Groups:         29        Scale:                1.8232
Min. group size:    3         Log-Likelihood:       -158.8308
Max. group size:    3         Converged:            Yes
Mean group size:    3.0
------------------------------------------------------------
                   Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept           3.425    0.194 17.632 0.000  3.045  3.806
questions_within_z -0.395    0.146 -2.712 0.007 -0.680 -0.109
groupID Var         0.487    0.265
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[40]:
```
model2_questions_between_sm = sm.MixedLM.from_formula(
    "cf ~ questions_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_questions_between_sm = model2_questions_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 questions between sm: CF ~ questions_between_z + (1|groupID)")
print(result2_questions_between_sm.summary())
```

```
Model 2 questions between sm: CF ~ questions_between_z + (1|groupID)
            Mixed Linear Model Regression Results
============================================================
Model:              MixedLM   Dependent Variable:   cf
No. Observations:   87        Method:               REML
No. Groups:         29        Scale:                2.0230
Min. group size:    3         Log-Likelihood:       -161.7841
Max. group size:    3         Converged:            Yes
Mean group size:    3.0
------------------------------------------------------------
                    Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept           3.425    0.196 17.481 0.000  3.041  3.809
questions_between_z 0.142    0.197  0.723 0.470 -0.244  0.529
groupID Var         0.439    0.258
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[41]:
```
model2_questions_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ questions_within_z + questions_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_questions_within_between_sm = model2_questions_within_between_sm.
 ↪fit(reml=True, method="lbfgs")
print("Model 2 questions within-between sm: CF ~ questions_within_z +␣
 ↪questions_between_z + (1|groupID)")
print(result2_questions_within_between_sm.summary())
```

```
Model 2 questions within-between sm: CF ~ questions_within_z +
questions_between_z + (1|groupID)
            Mixed Linear Model Regression Results
```

```
============================================================
Model:               MixedLM   Dependent Variable:   cf
No. Observations:    87        Method:               REML
No. Groups:          29        Scale:                1.8232
Min. group size:     3         Log-Likelihood:       -159.2768
Max. group size:     3         Converged:            Yes
Mean group size:     3.0
------------------------------------------------------------
                    Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept            3.425    0.196 17.481 0.000  3.041  3.809
questions_within_z  -0.395    0.146 -2.712 0.007 -0.680 -0.110
questions_between_z  0.142    0.197  0.723 0.470 -0.244  0.529
groupID Var          0.506    0.272
============================================================
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[42]:
```
model2_questions_within_bmb  = bmb.Model(
    "cf ~ questions_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_questions_within_bmb = model2_questions_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 questions within bmb: CF ~ questions_within_z + (1|groupID)")
az.summary(idata_model2_questions_within_bmb, var_names=["questions_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questions_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.

Model 2 questions within bmb: CF ~ questions_within_z + (1|groupID)
```

[42]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat
questions_within_z	-0.393	0.151	-0.63	-0.15	9982	5970	1.00
1|groupID_sigma	0.66	0.267	0.19	1.1	1639	2030	1.00
sigma	1.39	0.132	1.2	1.6	2843	3933	1.00

	mcse_mean	mcse_sd
questions_within_z	0.0015	0.0011
1|groupID_sigma	0.0066	0.0047
sigma	0.0025	0.0018

[43]:
```
model2_questions_between_bmb  = bmb.Model(
    "cf ~ questions_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_questions_between_bmb = model2_questions_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 questions between bmb: CF ~ questions_between_z + (1|groupID)")
az.summary(idata_model2_questions_between_bmb,␣
 ↪var_names=["questions_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questions_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.

Model 2 questions between bmb: CF ~ questions_between_z + (1|groupID)
```

[43]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat
questions_between_z	0.141	0.2	-0.17	0.46	5297	5307	1.00
1|groupID_sigma	0.62	0.283	0.13	1.1	1511	1564	1.00
sigma	1.458	0.134	1.3	1.7	3060	4667	1.00

```
                     mcse_mean mcse_sd
questions_between_z     0.0028  0.0021
1|groupID_sigma         0.0072  0.0049
sigma                   0.0024  0.0017
```

[44]:
```
model2_questions_within_between_bmb  = bmb.Model(
    "cf ~ questions_within_z + questions_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_questions_within_between_bmb = model2_questions_within_between_bmb.
 ↪fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 questions within-between bmb: CF ~ questions_within_z +␣
 ↪questions_between_z + (1|groupID)")
az.summary(idata_model2_questions_within_between_bmb,␣
 ↪var_names=["questions_within_z", "questions_between_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questions_within_z, questions_between_z,
1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 2 questions within-between bmb: CF ~ questions_within_z +
questions_between_z + (1|groupID)
```

[44]:
```
                       mean     sd eti89_lb eti89_ub  ess_bulk  ess_tail  \
questions_within_z   -0.395   0.15    -0.63    -0.15     12271      6191
questions_between_z   0.141  0.203    -0.18     0.47      4989      5387
1|groupID_sigma        0.67   0.27      0.2      1.1      1466      1533
sigma                 1.387  0.131      1.2      1.6      3354      4072
```

```
                    r_hat mcse_mean  mcse_sd
questions_within_z   1.00    0.0014  0.00098
questions_between_z  1.00    0.0029   0.0021
1|groupID_sigma      1.00    0.0071   0.0051
sigma                1.00    0.0023   0.0017
```

c) Pitch

1. Statsmodels (LME4)

[45]:
```
model2_pitch_within_sm = sm.MixedLM.from_formula(
    "cf ~ pitch_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
result2_pitch_within_sm = model2_pitch_within_sm.fit(reml=True, method="lbfgs")
print("Model 2 pitch within sm: CF ~ pitch_within_z + (1|groupID)")
print(result2_pitch_within_sm.summary())
```

```
Model 2 pitch within sm: CF ~ pitch_within_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   84      Method:             REML
No. Groups:         28      Scale:              1.9327
Min. group size:    3       Log-Likelihood:     -155.4275
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.417    0.201 16.987 0.000  3.022  3.811
pitch_within_z -0.166    0.150 -1.107 0.268 -0.461  0.128
groupID Var     0.488    0.271
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[46]:
```
model2_pitch_between_sm = sm.MixedLM.from_formula(
    "cf ~ pitch_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
```

```
)
result2_pitch_between_sm = model2_pitch_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 pitch between sm: CF ~ pitch_between_z + (1|groupID)")
print(result2_pitch_between_sm.summary())
```

```
Model 2 pitch between sm: CF ~ pitch_between_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   84      Method:             REML
No. Groups:         28      Scale:              1.9405
Min. group size:    3       Log-Likelihood:     -155.5600
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
                Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.411    0.204 16.713 0.000  3.011  3.811
pitch_between_z 0.115    0.210  0.550 0.582 -0.296  0.526
groupID Var     0.516    0.280
============================================================
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[47]:
```
model2_pitch_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ pitch_within_z + pitch_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
result2_pitch_within_between_sm = model2_pitch_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 pitch within-between sm: CF ~ pitch_within_z + pitch_between_z +␣
 ↪(1|groupID)")
print(result2_pitch_within_between_sm.summary())
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

```
Model 2 pitch within-between sm: CF ~ pitch_within_z + pitch_between_z +
(1|groupID)
          Mixed Linear Model Regression Results
```

```
============================================================
Model:              MixedLM  Dependent Variable:  cf
No. Observations:   84       Method:              REML
No. Groups:         28       Scale:               1.9327
Min. group size:    3        Log-Likelihood:      -155.9245
Max. group size:    3        Converged:           Yes
Mean group size:    3.0
------------------------------------------------------------
                Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept        3.411    0.204 16.713 0.000  3.011  3.811
pitch_within_z  -0.166    0.150 -1.107 0.268 -0.461  0.128
pitch_between_z  0.115    0.210  0.550 0.582 -0.296  0.526
groupID Var      0.518    0.281
============================================================
```

2. Bambi (Robustness testing)

[48]:
```
model2_pitch_within_bmb  = bmb.Model(
    "cf ~ pitch_within_z + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_model2_pitch_within_bmb = model2_pitch_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 pitch within bmb: CF ~ pitch_within_z + (1|groupID)")
az.summary(idata_model2_pitch_within_bmb, var_names=["pitch_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitch_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
```

```
Model 2 pitch within bmb: CF ~ pitch_within_z + (1|groupID)
```

[48]:
```
                    mean     sd eti89_lb eti89_ub  ess_bulk  ess_tail r_hat  \
pitch_within_z    -0.165  0.158    -0.42    0.091     11463      5494  1.00
1|groupID_sigma    0.67  0.278     0.18      1.1      1771      2439  1.00
sigma             1.427  0.139      1.2      1.7      3541      3526  1.00

                mcse_mean mcse_sd
pitch_within_z     0.0015  0.0011
1|groupID_sigma    0.0066  0.0045
sigma              0.0023  0.0017
```

[49]:
```
model2_pitch_between_bmb  = bmb.Model(
    "cf ~ pitch_between_z + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_model2_pitch_between_bmb = model2_pitch_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 pitch between bmb: CF ~ pitch_between_z + (1|groupID)")
az.summary(idata_model2_pitch_between_bmb, var_names=["pitch_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitch_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model 2 pitch between bmb: CF ~ pitch_between_z + (1|groupID)
```

[49]:
```
                    mean     sd eti89_lb eti89_ub  ess_bulk  ess_tail r_hat  \
pitch_between_z    0.112  0.219    -0.24     0.46      5770      5752  1.00
1|groupID_sigma     0.68   0.29     0.16      1.1      1433      1365  1.00
sigma              1.435  0.138      1.2      1.7      2902      4192  1.00

                mcse_mean mcse_sd
pitch_between_z     0.0029  0.0021
1|groupID_sigma     0.0077  0.0054
```

```
sigma              0.0026   0.002
```

[50]:
```
model2_pitch_within_between_bmb  = bmb.Model(
    "cf ~ pitch_within_z + pitch_between_z + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_model2_pitch_within_between_bmb = model2_pitch_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 pitch within-between bmb: CF ~ pitch_within_z + pitch_between_z␣
 ↪+ (1|groupID)")
az.summary(idata_model2_pitch_within_between_bmb, var_names=["pitch_within_z",␣
 ↪"pitch_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitch_within_z, pitch_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
```

```
Model 2 pitch within-between bmb: CF ~ pitch_within_z + pitch_between_z +
(1|groupID)
```

[50]:
```
                    mean     sd eti89_lb eti89_ub  ess_bulk  ess_tail r_hat  \
pitch_within_z    -0.165  0.157    -0.41    0.087      8387      5160  1.00
pitch_between_z    0.113  0.217    -0.23     0.46      4567      4617  1.00
1|groupID_sigma     0.68  0.282     0.18      1.1      1767      2225  1.00
sigma              1.434  0.141      1.2      1.7      2869      3356  1.00

                  mcse_mean mcse_sd
pitch_within_z       0.0017  0.0012
pitch_between_z      0.0032  0.0025
1|groupID_sigma      0.0067  0.0047
sigma                0.0027   0.002
```

1.2.3 Model 3 - Performance Perception: CF ~ pp_within/between_z + (1|groupID)

1. Statsmodels (LME4)

[51]:
```
model3_pp_within_sm = sm.MixedLM.from_formula(
    "cf ~ pp_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result3_pp_within_sm = model3_pp_within_sm.fit(reml=True, method="lbfgs")
print("Model 3 pp within sm: CF ~ pp_within_z + (1|groupID)")
print(result3_pp_within_sm.summary())
```

```
Model 3 pp within sm: CF ~ pp_within_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              2.0545
Min. group size:   3       Log-Likelihood:     -162.2339
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     3.425    0.194 17.632 0.000  3.045  3.806
pp_within_z   0.052    0.155  0.334 0.738 -0.251  0.355
groupID Var   0.410    0.249
=========================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[52]:
```
model3_pp_between_sm = sm.MixedLM.from_formula(
    "cf ~ pp_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result3_pp_between_sm = model3_pp_between_sm.fit(reml=True, method="lbfgs")
print("Model 3 pp between sm: CF ~ pp_between_z + (1|groupID)")
print(result3_pp_between_sm.summary())
```

```
Model 3 pp between sm: CF ~ pp_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
```

```
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              2.0230
Min. group size:  3       Log-Likelihood:     -156.4649
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------
Intercept     3.425    0.161 21.288 0.000  3.110  3.741
pp_between_z  0.601    0.162  3.717 0.000  0.284  0.919
groupID Var   0.076    0.174
======================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[53]:
```
model3_pp_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ pp_within_z + pp_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result3_pp_within_between_sm = model3_pp_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 3 pp within-between sm: CF ~ pp_within_z + pp_between_z +␣
 ↪(1|groupID)")
print(result3_pp_within_between_sm.summary())
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)

Model 3 pp within-between sm: CF ~ pp_within_z + pp_between_z + (1|groupID)
         Mixed Linear Model Regression Results
======================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              2.0545
Min. group size:  3       Log-Likelihood:     -157.3608
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------
```

```
Intercept     3.425    0.161 21.288 0.000  3.110  3.741
pp_within_z   0.052    0.155  0.334 0.738 -0.251  0.355
pp_between_z  0.601    0.162  3.717 0.000  0.284  0.919
groupID Var   0.066    0.173
============================================================
```

2. Bambi (Robustness testing)

[54]:
```
model3_pp_within_bmb  = bmb.Model(
    "cf ~ pp_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model3_pp_within_bmb = model3_pp_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 3 pp within bmb: CF ~ pp_within_z + (1|groupID)")
az.summary(idata_model3_pp_within_bmb, var_names=["pp_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pp_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 3 pp within bmb: CF ~ pp_within_z + (1|groupID)
```

[54]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
pp_within_z	0.051	0.16	-0.2	0.31	10477	6148	1.00	
1|groupID_sigma	0.6	0.281	0.12	1	1597	2047	1.00	
sigma	1.47	0.134	1.3	1.7	3416	4511	1.00	

	mcse_mean	mcse_sd
pp_within_z	0.0016	0.0011
1|groupID_sigma	0.007	0.0047
sigma	0.0023	0.0017

[55]:
```
model3_pp_between_bmb  = bmb.Model(
    "cf ~ pp_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model3_pp_between_bmb = model3_pp_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 3 pp between bmb: CF ~ pp_between_z + (1|groupID)")
az.summary(idata_model3_pp_between_bmb, var_names=["pp_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pp_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model 3 pp between bmb: CF ~ pp_between_z + (1|groupID)
```

[55]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
pp_between_z	0.599	0.174	0.33	0.88	6738	5835	1.00	
1|groupID_sigma	0.36	0.229	0.038	0.75	1761	2535	1.00	
sigma	1.424	0.119	1.2	1.6	5328	5161	1.00	

	mcse_mean	mcse_sd
pp_between_z	0.0021	0.0016
1|groupID_sigma	0.0051	0.0035
sigma	0.0016	0.0012

[56]:
```
model3_pp_within_between_bmb  = bmb.Model(
    "cf ~ pp_within_z + pp_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model3_pp_within_between_bmb = model3_pp_within_between_bmb.fit(
    draws=2000,
```

```
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 3 pp within-between bmb: CF ~ pp_within_z + pp_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model3_pp_within_between_bmb, var_names=["pp_within_z",␣
 ↪"pp_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pp_within_z, pp_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model 3 pp within-between bmb: CF ~ pp_within_z + pp_between_z + (1|groupID)
```

[56]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
pp_within_z	0.053	0.155	-0.2	0.3	13155	6272	1.00	
pp_between_z	0.602	0.173	0.33	0.88	9215	6051	1.00	
1|groupID_sigma	0.346	0.227	0.034	0.74	2136	2724	1.00	
sigma	1.434	0.119	1.3	1.6	5596	5626	1.00	

	mcse_mean	mcse_sd
pp_within_z	0.0014	0.00095
pp_between_z	0.0018	0.0013
1|groupID_sigma	0.0047	0.0032
sigma	0.0016	0.0012

1.2.4 Model 4 - Uncertainty Score: CF ~ uncertScore + (1|groupID)

1. Statsmodels (LME4)

[57]:
```
model4_sm = sm.MixedLM.from_formula(
    "cf ~ uncertScore_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result4_sm = model4_sm.fit(reml=True, method="lbfgs")
print("Model 4 sm: CF ~ uncertScore_z + (1|groupID)")
```

```
print(result4_sm.summary())
```

```
Model 4 sm: CF ~ uncertScore_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              2.0230
Min. group size:   3       Log-Likelihood:     -162.0425
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
              Coef. Std.Err.    z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     3.425    0.198 17.315 0.000  3.038  3.813
uncertScore_z 0.005    0.199  0.024 0.981 -0.385  0.395
groupID Var   0.461    0.263
=========================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[58]:
```
model4_bmb  = bmb.Model(
    "cf ~ uncertScore_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model4_bmb = model4_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 4 bmb: CF ~ uncertScore_z + (1|groupID)")
az.summary(idata_model4_bmb, var_names=["uncertScore_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, uncertScore_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model 4 bmb: CF ~ uncertScore_z + (1|groupID)
```

[58]:
```
                   mean     sd eti89_lb eti89_ub  ess_bulk  ess_tail r_hat  \
uncertScore_z     0.008  0.204    -0.31     0.34      5738      5586  1.00
1|groupID_sigma    0.65  0.277     0.17      1.1      1533      1366  1.00
sigma             1.455  0.133      1.3      1.7      3469      3945  1.00

                 mcse_mean mcse_sd
uncertScore_z       0.0027   0.002
1|groupID_sigma      0.007  0.0048
sigma               0.0023  0.0017
```

1.2.5 Follow-up models

Model Follow-up 1 Questions Path: CF ~ questionPath + (1|groupID)

1. Statsmodels (LME4)

[59]:
```
modelFU1_sm = sm.MixedLM.from_formula(
    "cf ~ questionPath",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
resultFU1_sm = modelFU1_sm.fit(reml=True, method="lbfgs")
print("Model FU1 sm: CF ~ questionPath + (1|groupID)")
print(resultFU1_sm.summary())
```

```
Model FU1 sm: CF ~ questionPath + (1|groupID)
           Mixed Linear Model Regression Results
============================================================
Model:               MixedLM  Dependent Variable:  cf
No. Observations:    87       Method:              REML
No. Groups:          29       Scale:               2.0230
Min. group size:     3        Log-Likelihood:      -161.1184
Max. group size:     3        Converged:           Yes
Mean group size:     3.0
------------------------------------------------------------
                  Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept          3.500    0.237 14.779 0.000  3.036  3.964
questionPath[T.1] -0.241    0.425 -0.566 0.571 -1.074  0.592
groupID Var        0.447    0.260
```

```
============================================================
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[60]:
```
modelFU1_bmb  = bmb.Model(
    "cf ~ questionPath + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_modelFU1_bmb = modelFU1_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model FU1 bmb: CF ~ questionPath + (1|groupID)")
az.summary(idata_modelFU1_bmb, var_names=["questionPath", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questionPath, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model FU1 bmb: CF ~ questionPath + (1|groupID)
```

[60]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
questionPath[1]	-0.25	0.439	-0.95	0.45	6070	5460	1.00	
1|groupID_sigma	0.63	0.29	0.13	1.1	1094	967	1.00	
sigma	1.458	0.134	1.3	1.7	3191	4209	1.00	

	mcse_mean	mcse_sd
questionPath[1]	0.0056	0.0041
1|groupID_sigma	0.0086	0.0058
sigma	0.0024	0.0017

Model Follow-up 2 PitchRole: CF ~ pitchRole + (1|groupID)

1. Statsmodels (LME4)

[61]:
```
modelFU2_sm = sm.MixedLM.from_formula(
    "cf ~ pitchRole",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
resultFU2_sm = modelFU2_sm.fit(reml=True, method="lbfgs")
print("Model FU2 sm: CF ~ pitchRole + (1|groupID)")
print(resultFU2_sm.summary())
```

```
Model FU2 sm: CF ~ pitchRole + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   84      Method:             REML
No. Groups:         28      Scale:              1.8873
Min. group size:    3       Log-Likelihood:     -153.6994
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.735    0.268 13.937 0.000  3.210  4.260
pitchRole[T.1] -0.581    0.328 -1.774 0.076 -1.224  0.061
groupID Var     0.480    0.268
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[62]:
```
modelFU2_bmb  = bmb.Model(
    "cf ~ pitchRole + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_modelFU2_bmb = modelFU2_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
```

```
    cores=4,
    random_seed=123
)

print("Model FU2 bmb: CF ~ pitchRole + (1|groupID)")
az.summary(idata_modelFU2_bmb, var_names=["pitchRole", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitchRole, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
```

```
Model FU2 bmb: CF ~ pitchRole + (1|groupID)
```

[62]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
pitchRole[1]	-0.589	0.338	-1.1	-0.053	7342	5596	1.00	
1|groupID_sigma	0.66	0.28	0.17	1.1	1506	1845	1.00	
sigma	1.408	0.137	1.2	1.6	2715	3478	1.00	

	mcse_mean	mcse_sd
pitchRole[1]	0.0039	0.0028
1|groupID_sigma	0.0072	0.005
sigma	0.0026	0.002